Daniel S.W. Lee, Ph.D.

ORCID iD: orcid.org/0000-0002-4690-9012 • Email: dswlee@berkeley.edu • Website: danswlee.github.io

EDUCATION

PRINCETON UNIVERSITY — Princeton, NJ
PhD, Quantitative and Computational Biology — 2021

PRINCETON UNIVERSITY — Princeton, NJ
MA, Quantitative and Computational Biology — 2018

YALE UNIVERSITY — New Haven, CT
BS, Physics; Molecular Biophysics and Biochemistry — 2016

RESEARCH EXPERIENCE

CALIFORNIA INSTITUTE FOR QUANTITATIVE BIOSCIENCES, UC BERKELEY — Berkeley, CA
Postdoctoral Scholar|Advisor: Prof. Dan Fletcher — Apr. 2022 – Present

LEWIS-SIGLER INSTITUTE FOR INTEGRATIVE GENOMICS, PRINCETON UNIVERSITY — Princeton, NJ
Postdoctoral Research Associate|Advisor: Prof. Ned Wingreen — Nov. 2021 – Apr. 2022

LEWIS-SIGLER INSTITUTE FOR INTEGRATIVE GENOMICS, PRINCETON UNIVERSITY — Princeton, NJ
PhD Student|Advisors: Profs. Cliff Brangwynne and Ned Wingreen — Sept. 2016 – Nov. 2021

DEPT. OF BIOSTATISTICS, YALE SCHOOL OF PUBLIC HEALTH — New Haven, CT
Research Assistant|Advisor: Prof. Jeffrey Townsend — Jan. 2016 – Aug. 2016

DEPT. OF MOLECULAR BIOPHYSICS AND BIOCHEMISTRY, YALE UNIVERSITY — New Haven, CT
Undergraduate Research Student|Advisor: Prof. Jonathon Howard — Aug. 2014 – Aug. 2016

SCHOOL OF LIFE SCIENCES, TSINGHUA UNIVERSITY — Beijing, China
Research Assistant|Advisor: Prof. Xin Liang — May 2015 – Aug. 2015

MAX PLANCK INSTITUTE FOR CELL BIOLOGY & GENETICS — Dresden, Germany
Undergraduate Research Student|Advisor: Prof. Jonathon Howard — May 2014 – Aug. 2014

DEPT. OF MOLECULAR BIOPHYSICS AND BIOCHEMISTRY, YALE UNIVERSITY — New Haven, CT
Research Assistant|Advisor: Prof. Patrick Sung — May 2013 – May 2014

AWARDS, HONORS & FUNDING

BURROUGHS WELLCOME FUND
Career Award at the Scientific Interface — July 2026 – July 2031

NATIONAL SCIENCE FOUNDATION GRADUATE RESEARCH FELLOW
NSF-GRFP in Physics and Astronomy, Physics of Living Systems — Sept. 2018 – May 2021

PUBLICATIONS

2,494 citations, *h*=13 (from **Google Scholar**, current as of 08/18/2026)
* indicates equal contribution
† indicates corresponding author

Lee, D.S.W.*, Oster, L.F.*, Son, S., Fletcher, D.A.† **Cell surface crowding is a tunable energetic barrier to cell-cell fusion**. *Nature Communications* **16**(1), 7158 (2025).

Choi, C.-H., **Lee, D.S.W.**, Sanders, D.W., Brangwynne, C.P.† **Condensate interfaces can accelerate protein aggregation**. *Biophysical Journal* **123**(11), 1404-1413 (2024).

Takatori, S.C.*†, Son, S.*†, **Lee, D.S.W.**, Fletcher, D.A.† **Engineered molecular sensors for quantifying cell surface crowding**. *Proceedings of the National Academy of Sciences* **120**(21), e2219778120 (2023).

Lee, D.S.W., Choi, C.-H., Sanders, D.W., Beckers, L., Riback, J.A., Brangwynne, C.P.†, Wingreen, N.S.† **Size distributions of intracellular condensates reflect competition between coalescence and nucleation**. *Nature Physics* **19**(4), 586-596 (2023).

Riback, J.A.*, Eeftens, J.M.*, **Lee, D.S.W.***, Quinodoz, S.A., Donlic, A., Orlovsky, N., Wiesner, L., Beckers, L., Becker, L.A., Strom, A.R., Rana, U., Tolbert, M., Purse, B.W., Kleiner, R., Kriwacki, R., Brangwynne, C.P.† **Viscoelasticity and advective flow of RNA underlies nucleolar form and function**. *Molecular Cell* **83**(17), 3095-3107.e9 (2023).

Lee, D.S.W., Strom, A.R., Brangwynne, C.P.† **The mechanobiology of nuclear phase separation**. *APL Bioengineering* **6**(2), 021503 (review, 2022).

Jack, A.*, Kim, Y.*, Strom, A.R.*, **Lee, D.S.W.**, Williams, B., Schaub, J.M., Kellogg, E.H., Finkelstein, I.J., Ferro, L.S.†, Yildiz, A.†, Brangwynne, C.P.† **Compartmentalization of telomeres through DNA-scaffolded phase separation**. *Developmental Cell* **57**(2), 277-290.e9 (2022).

Zhang, Y.*, **Lee, D.S.W.***, Meir, Y., Brangwynne, C.P., Wingreen, N.S.† **Mechanical frustration of phase separation in the cell nucleus by chromatin**. *Physical Review Letters* **126**(25), 258102 (2021).

Lin, C.K.*, **Lee, D.S.W.***, McKeithen-Mead, S., Emonet, T., Kazmierczak, B.† **A primed subpopulation of bacteria allows rapid expression of the T3SS in** *Pseudomonas aeruginosa*. *mBio* **12**(3), e00831-21 (2021).

Lee, D.S.W., Wingreen, N.S.†, Brangwynne, C.P.† **Chromatin mechanics dictates subdiffusion and coarsening dynamics of embedded condensates**. *Nature Physics* **17**(4), 531-538 (2021). Featured on April 2021 cover.

Sanders, D.W., Kedersha, N.*, **Lee, D.S.W.***, Strom, A.R.*, Drake, V., Riback, J.A., Bracha, D., Eeftens, J.M., Iwanicki, A., Wang, A., Wei, M.T., Whitney, G., Lyons, S.M., Anderson, P., Jacobs, W.M., Ivanov, P., Brangwynne, C.P.† **Competing protein-RNA interaction networks control multiphase intracellular organization**. *Cell* **181**(2), 306-324.e28 (2020).

Shin, Y.*, Chang, Y.-C.*, **Lee, D.S.W.**, Berry, J., Sanders, D.W., Ronceray, P., Wingreen, N.S., Haataja, M., Brangwynne, C.P.† **Liquid nuclear condensates mechanically sense and restructure the genome**. *Cell* **175**(6), 1481-1491.e13 (2018).

Zhao, Z.M.*, Campbell, M.C.*, Li, N., **Lee, D.S.W.**, Zhang, Z., Townsend, J.P.† **Detection of regional variation in selection intensity within protein-coding genes using DNA sequence polymorphism and divergence**. *Molecular Biology and Evolution* **34**(11), 3006-3022 (2017).

PROFESSIONAL ACTIVITIES, TEACHING & SERVICE

JOURNAL REFEREE 2024 – Present
Reviewed manuscripts for *PRL*, *PRE*, and *PRX Life*

CLAUDE SCIENCE COHORT Sept. – Dec. 2026
Selected as 1 of 50 teams for the inaugural Claude Science Cohort to receive in-kind support ($30,000 in Claude API credits and $2,000 in Modal compute) for project "Discovering strategies to manipulate cell surface accessibility" using AI to link gene expression to cell surface biophysics.

MENTORSHIP
Mentored undergraduate Alan Amin (summer 2018) in project investigating mechanical perturbation of chromatin in living cells, undergraduate Angela Yang (summer 2020) in developing and analyzing polymer simulations in LAMMPS molecular dynamics package, Molecular Biology rotation student Jordy Botello in high-throughput image analysis project (2021), and undergraduate Natalia Orlovsky in project on subdiffusive fluctuations in the nucleus (2021).

Mentored Fletcher lab rotation students Maddie Blaauw and Kira Buttrey (UCB BioE program) in developing oligo-based cell surface crowding probes; undergraduate Yuri Onimura (Molecular Cell Biology major) in project analyzing CRISPR screening readouts; intern Yue See Lee (California Institute for Regenerative Medicine program through Berkeley City College) in project on myoblast fusion.

MCGRAW TEACHING TRANSCRIPT
Participated in McGraw teaching transcript program, focusing on training pedagogical skills. Activities include attending workshops, designing teaching statement and sample syllabus and receiving feedback on class observations.

WOODS HOLE PHYSIOLOGY COURSE Summer 2019, 2023, 2024
Teaching Assistant. Designed and administered tutorials and independent projects for students in Woods Hole Physiology Course 2019 as part of Brangwynne lab and in 2023 and 2024 as part of Fletcher lab.

ASSISTANT IN INSTRUCTION Fall 2018, 2019
QCB 515, Methods and Logic in Quantitative and Computational Biology. Designed problems and lessons to instruct interdisciplinary group of students in computational and theoretical approaches to biology under Prof. Ned Wingreen.

2017 CARGÈSE SUMMER SCHOOL ON THEORETICAL BIOPHYSICS June – July 2017
Student
Participated in two-week summer school designed to instruct graduate students in theoretical methods and approaches to biophysics held in Corsica, France, culminating in original research projects. Supervised by Thierry Mora, Aleksandra Walczak, and Ilya Nemenman.

YALE UNDERGRADUATE RESEARCH ASSOCIATION Jan. 2014 – May 2016
Founder & Director
Founded organization to build and serve community among undergraduates pursuing research at Yale.

CONFERENCES & TALKS

Lee, D.S.W., **Regulation and manipulation of immune cell surface crowding**. Invited Seminar, Rice University. **Future Leaders in Bioengineering Seminar Series**, Nov. 2026.

Lee, D.S.W., Oster, L.F., Son, S., Fletcher, D.A. **Cell Surface Crowding is a Biophysical Barrier to Cell-Cell Fusion**. **American Society for Cell Biology 2024 Meeting**, Subgroup Talk.

Lee, D.S.W., Kim, J.Y., Robey, E.A., Fletcher, D.A. **Regulation of Cell Surface Crowding on Cancer Cells and Immune Cells**. **American Society for Cell Biology 2023 Meeting**, Subgroup Talk.

Lee, D.S.W., Brangwynne, C.P., Wingreen, N.S. **Physical Principles Underlying Intracellular Condensate Size Distributions**. Invited Seminar, **NYU Chromatin Club**, September 29, 2023.

Lee, D.S.W., Brangwynne, C.P., Wingreen, N.S. **Physical Principles Underlying Intracellular Condensate Size Distributions**. **APS March Meeting 2022**.

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Chromatin Mechanics Dictates Subdiffusion and Coarsening Dynamics of Embedded Condensates**. **American Society for Cell Biology 2020 Virtual Meeting**, Subgroup Talk.

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Subdiffusive Dynamics of Optogenetic Droplets Measure Chromatin Mechanics**. Invited Talk, **NSF-Funded MIT Genome Architecture and Function Virtual Workshop 2020**.

Lee, D.S.W., Chang, Y.-C., Shin, Y., Sanders, D.W., Bracha, D., Wingreen, N.S., Brangwynne, C.P. **Mechanical Interplay of Chromatin and Liquid-Liquid Phase Separated Condensates**. **APS March Meeting 2019**, Boston, MA.

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Subdiffusive Dynamics of Optogenetic Droplets Measure Chromatin Mechanics**. **2018 4DN-ASCB Satellite Meeting**, San Diego, CA, Lightning Talk & Poster.

Lee, D.S.W., Wingreen, N.S., Brangwynne, C.P. **Dynamic Organization of Chromatin by Liquid-Liquid Phase Separation**. **2018 National Human Genome Research Institute Training and Career Development Meeting**, Los Angeles, CA, Lightning Talk.